

TC

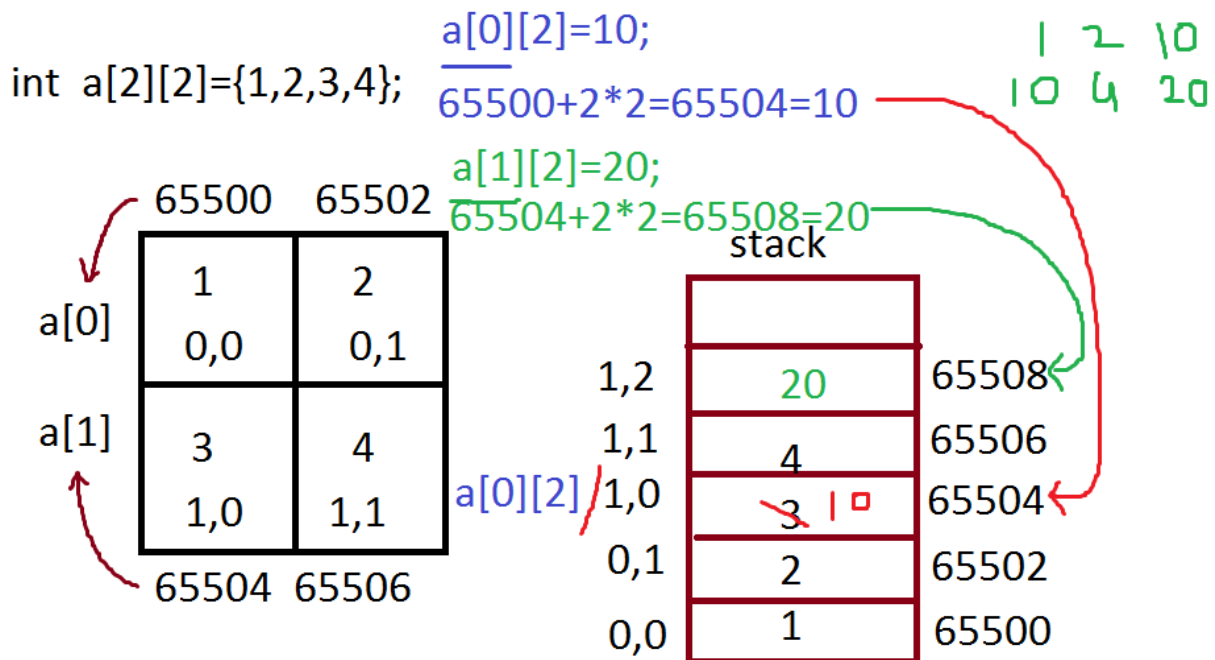
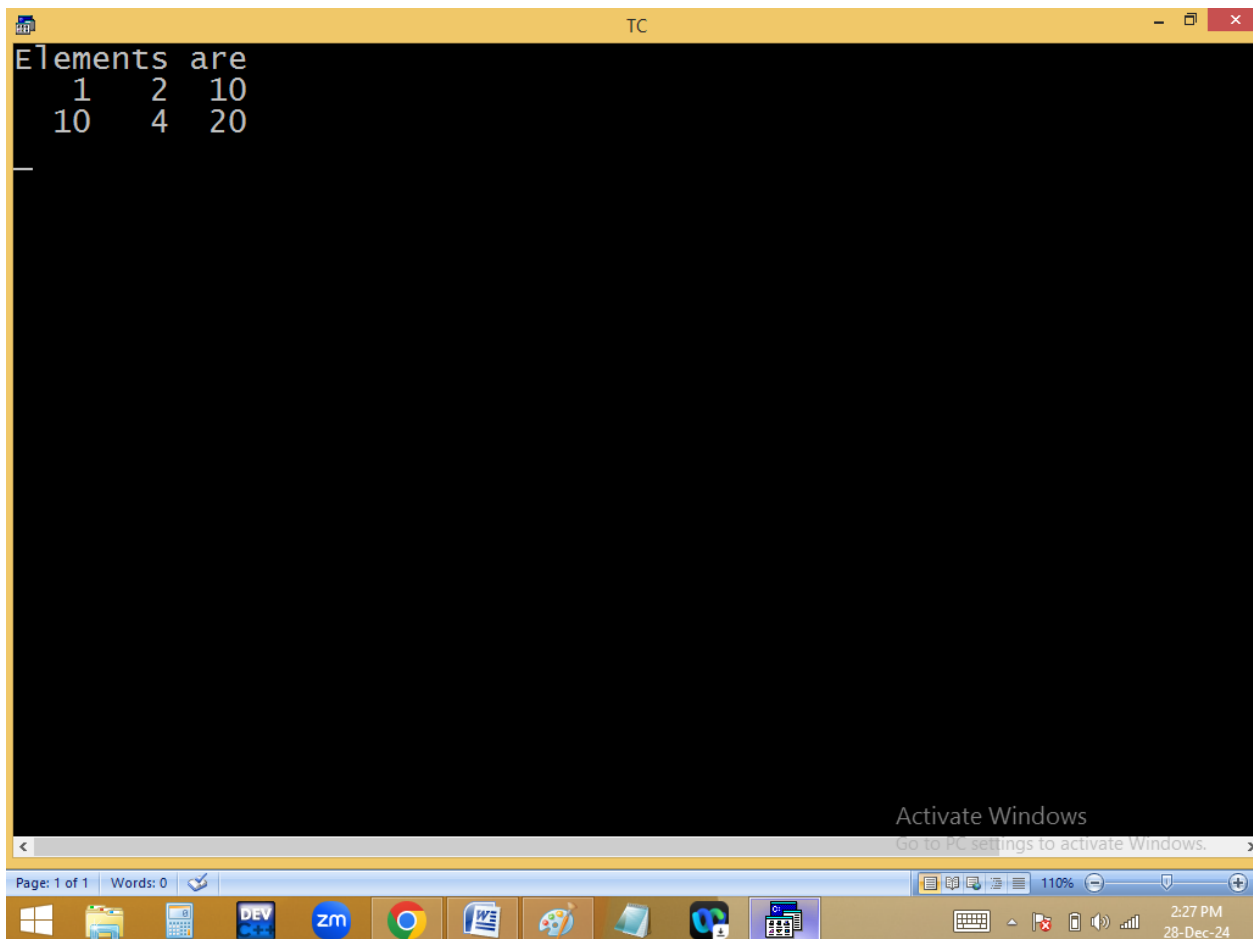
File Edit Run Compile Project Options Debug Breakpoint
Line 1 Col 1 Insert Indent Tab Fill Unindent E:2PM

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[2][2]={1,2,3,4},r,c;
a[0][2]=10;
a[1][2]=20;
clrscr();
puts("Elements are ");
for(r=0;r<2;r++)
{
for(c=0;c<3;c++)
{
printf("%4d",a[r][c]);
}
printf("\n");
}
getch();
}
```

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Me
Go to PC settings to activate Windows.

Build 9600

2:26 PM
28-Dec-24



Reading and printing elements of $n \times n$ matrix:

```
TC
File Edit Run Compile Project Options Debug Break
Line 13 Col 13 Insert Indent Tab Fill Unindent * E:2PM
#include<stdio.h>
#include<conio.h>
void main()
{
int a[10][10],nr,nc,r,c;
clrscr();
printf("Enter no of rows and columns ");scanf("%d%d",&nr,&nc);
printf("Enter %d integers\n",nr*nc);
for(r=0;r<nr;r++)for(c=0;c<nc;c++)scanf("%d",&a[r][c]);
puts("Elements are ");
for(r=0;r<nr;r++)
{
for(c=0;c<nc;c++)
{
printf("%4d",a[r][c]);
}
printf("\n");
}
getch();
}
```

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Me

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Page: 3 of 3 Words: 7 110%

2:29 PM 28-Dec-24

```
TC
Enter no of rows and columns 3 3
Enter 9 integers
1 2 3 4 5 6 7 8 9
Elements are
  1  2  3
  4  5  6
  7  8  9
```

Activate Windows
Go to PC settings to activate Windows.

Page: 3 of 3 Words: 7

110%

2:30 PM
28-Dec-24

```
TC
Enter no of rows and columns 5 2
Enter 10 integers
2 0 1 7 3 -7 8 3 9 0
Elements are
  2  0
  1  7
  3 -7
  8  3
  9  0
```

Activate Windows
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Page: 5 of 5 Words: 7

2:30 PM
28-Dec-24

Transpose of $n \times n$ matrix:

```
TC
File Edit Run Compile Project Options Debug Bre
Line 10 Col 18 Insert Indent Tab Fill Unindent * E:2PM
#include<stdio.h>
#include<conio.h>
void main()
{
int a[10][10],nr,nc,r,c;
clrscr();
printf("Enter no of rows and columns ");scanf("%d%d",&nr,&nc);
printf("Enter %d integers\n",nr*nc);
for(r=0;r<nr;r++)for(c=0;c<nc;c++)scanf("%d",&a[r][c]);
puts("Transposed Elements are ");
for(c=0;c<nc;c++)
{
for(r=0;r<nr;r++)
{
printf("%4d",a[r][c]);
}
printf("\n");
}
getch();
}
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Me
Go to PC settings to activate Windows.
1894 x 763px 100%
Windows taskbar with icons for File Explorer, Calculator, DEV, zm, Chrome, Word, Paint, and others. System tray shows 2:36 PM on 28-Dec-24.
```

```

TC
Enter no of rows and columns 2 3
Enter 6 integers
1 2 3
4 5 6
Transposed Elements are
1 4
2 5
3 6

```

Activate Windows
Go to PC settings to activate Windows.

Page: 7 of 7 Words: 11 110%

2:36 PM 28-Dec-24

```

for( c=0; c<3; c++) ✓
{
    for( r=0; r<2; r++)
    {
        p(a[r][c]);
    }
    p("\n");
}

```

1 0,0	2 0,1	3 0,2
4 1,0	5 1,1	6 1,2

1 0,0	4 0,1
2 1,0	5 1,1
3 2,0	6 2,1

$\frac{r}{0} \mid \cancel{1} \mid \frac{c}{0}$
 $0 \mid \cancel{2} \mid 1$
 $0 \mid \cancel{2} \mid 2$
 $\quad \quad \quad \cancel{3}$

Method 2:

```
TC
File Edit Run Compile Project Options Debug Bre
Line 15 Col 1 Insert Indent Tab Fill Unindent * E:2PM
#include<stdio.h>
#include<conio.h>
void main()
{
int a[10][10],nr,nc,r,c;
clrscr();
printf("Enter no of rows and columns ");scanf("%d%d",&nr,&nc);
printf("Enter %d integers\n",nr*nc);
for(r=0;r<nr;r++)for(c=0;c<nc;c++)scanf("%d",&a[r][c]);
puts("Transposed Elements are ");
for(r=0;r<nc;r++)
{
for(c=0;c<nr;c++)
{
printf("%4d",a[c][r]);
}
printf("\n");
}
getch();
}
```

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Me

Go to PC settings to activate Windows.

1894 x 763px 100%

2:40 PM 28-Dec-24

```

TC
Enter no of rows and columns 4 2
Enter 8 integers
1 2
3 4
5 6
7 8
Transposed Elements are
  1  3  5  7
  2  4  6  8

```

Activate Windows
Go to PC settings to activate Windows.

Page: 9 of 10 Words: 13 110%

2:40 PM 28-Dec-24

```

for( r=0; r<3; r++)
{
  for( c=0; c<2; c++)
  {
    p(a[c][r]);
  }
  p("\n");
}

```

¹ 0,0	² 0,1	³ 0,2
⁴ 1,0	⁵ 1,1	⁶ 1,2

✓ ¹ 0,0	✓ ⁴ 0,1
✓ ² 1,0	✓ ⁵ 1,1
✓ ³ 2,0	✓ ⁶ 2,1

C r
0 1 ~~2~~ 0
0 1 ~~2~~ 1
0 1 ~~2~~ 2
 ~~3~~

Finding even/odd/zero elements row wise:

```
TC
File Edit Run Compile Project Options Debug Break
Line 14 Col 19 Insert Indent Tab Fill Unindent * E:2PM
#include<stdio.h>
#include<conio.h>
void main()
{
int a[10][10],nr,nc,r,c,e,o,z;
clrscr();
printf("Enter no of rows and columns ");scanf("%d%d",&nr,&nc);
printf("Enter %d integers\n",nr*nc);
for(r=0;r<nr;r++)for(c=0;c<nc;c++)scanf("%d",&a[r][c]);
puts("\t Even\todd\tZero");
puts("-----");
for(r=0;r<nr;r++)
{
for(e=o=z=c=0;c<nc;c++)
{
if(a[r][c]==0)z++;else if(a[r][c]%2==0)e++;else o++;
}
printf("%d-row\t %d\t%d\t%d\n",r+1,e,o,z);
}
getch();
}
```

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Me

Go to PC settings to activate Windows.

1894 x 763px 100%

2:59 PM 28-Dec-24

```

Enter no of rows and columns 3 4
Enter 12 integers
1 2 3 0
4 5 6 7
0 8 9 0

Even    odd    Zero
-----
1-row   1      2      1
2-row   2      2      0
3-row   1      1      2

```

```
puts("\t Even\tOdd\tZero");
puts("-----");
```

```
for(r=0;r<2;r++)
{
for(e=o=z=c=0;c<4;c++)
```

```

{
if(a[r][c]==0)z++;
else if(a[r][c]%2==0)e++;
else o++;
}
p("%d-row\t%d\t%d\t%d\n",r+1,e,o,z);
}

```

1 0,0	2 0,1	3 0,2	0 0,2
4 1,0	5 1,1	6 1,2	7 1,2

	Even	Odd	Zero
1-row	1	2	1
2-row	2	2	0

Column wise:

```
TC
File Edit Run Compile Project Options Debug Bre
Line 16 Col 38 Insert Indent Tab Fill Unindent * E:2PM
#include<stdio.h>
#include<conio.h>
void main()
{
int a[10][10],nr,nc,r,c,e,o,z;
clrscr();
printf("Enter no of rows and columns ");scanf("%d%d",&nr,&nc);
printf("Enter %d integers\n",nr*nc);
for(r=0;r<nr;r++)for(c=0;c<nc;c++)scanf("%d",&a[r][c]);
puts("\t Even\todd\tZero");
puts("-----");
for(c=0;c<nc;c++)
{
for(e=o=z=r=0;r<nr;r++)
{
if(a[r][c]==0)z++;else if(a[r][c]%2==0)e++;else o++;
}
printf("%d-col\t %d\t%d\t%d\n",c+1,e,o,z);
}
getch();
}

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Me
Go to PC settings to activate Windows.
1894 x 768px 100%
3:12 PM
28-Dec-24
```

```
TC
Enter no of rows and columns 3 4
Enter 12 integers
1 2 3 0
4 5 6 7
0 8 0 9
Even    odd    Zero
-----
1-col   1      1      1
2-col   2      1      0
3-col   1      1      1
4-col   0      2      1
_
```

Activate Windows
Go to PC settings to activate Windows.

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row sum and column sum

```
#include<stdio.h>
```

```
#include<conio.h>
```

```
void main()
```

```
{
```

```
int a[10][10],nr,nc,r,c,rs,cs;
```

```
clrscr();
```

```
printf("Enter no of rows ");scanf("%d",&nr);
```

```
printf("Enter %d integers\n",nr*nr);
for(r=0;r<nr;r++)for(c=0;c<nr;c++)scanf("%d",&a[r][c]);

for(r=0;r<nr;r++)
{
for(c=rs=cs=0;c<nr;c++)
{rs+=a[r][c]; cs+=a[c][r];}
a[r][c]=rs;a[c][r]=cs;
}

puts("Result elements are ");
puts("-----");
for(r=0;r<=nr;r++)
{
for(c=0;c<=nr;c++)
{
if(r==nr&& c==nr); else printf("%4d",a[r][c]);
```

```
}
```

```
printf("\n");
```

```
}
```

```
getch();
```

```
}
```

```

Enter no of rows 3
Enter 9 integers
1 2 3
4 5 6
7 8 9
Result elements are
-----
1 2 3 6
4 5 6 15
7 8 9 24
12 15 18

```

```

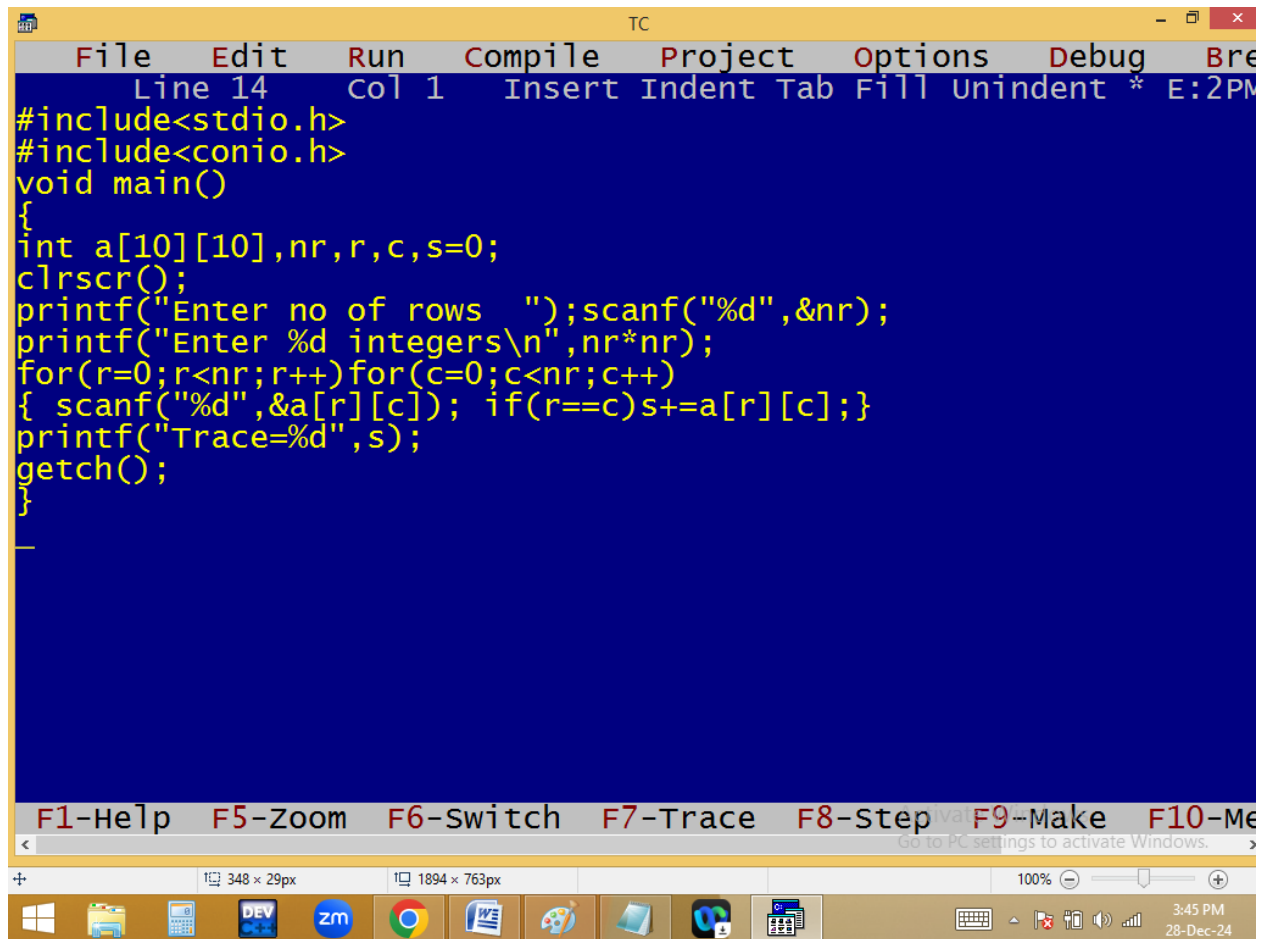
for( r=0; r<2; r++ )
{
  for( rs=cs=c=0; c<2; c++ ) ✓
  {
    rs+=a[r][c]; cs+=a[c][r];
  }
  a[r][c]=rs; a[c][r]=cs;
}

```

1 ✓ 0,0	2 0,1	3 0,2
4 1,0	5 1,1	9 ✓ 1,2
5 2,0	7 ✓ 2,1	2 2,2

r	c	rs	cs
0	0	0	0
		1	1
		2	4
		3	5
1	0	0+4+5=9	
	1	0+2+5=7	

Finding trace of n*n matrix:



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break. The status bar at the top indicates "Line 14 Col 1 Insert Indent Tab Fill Unindent * E:2PM". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[10][10],nr,r,c,s=0;
clrscr();
printf("Enter no of rows ");scanf("%d",&nr);
printf("Enter %d integers\n",nr*nr);
for(r=0;r<nr;r++)for(c=0;c<nr;c++)
{ scanf("%d",&a[r][c]); if(r==c)s+=a[r][c];}
printf("Trace=%d",s);
getch();
}
```

Below the code, there is a row of function key shortcuts: F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, and F10-Me. A watermark "WUOLAH" is visible in the background. The bottom status bar shows "Go to PC settings to activate Windows." and a taskbar with various application icons. The system clock in the bottom right corner displays "3:45 PM 28-Dec-24".

```
TC
Enter no of rows 3
Enter 9 integers
1 0 7
2 8 1
4 7 6
Trace=15
```

Activate Windows
Go to PC settings to activate Windows.

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Trace: Sum of principle diagonal elements

if(r==c) s+=a[r][c];

1 ✓ 0,0	2 0,1	3 0,2
4 1,0	5 1,1	6 1,2
7 2,0	8 ✓ 2,1	9 2,2

trace =14

Finding trace of right diagonal elements

The image shows a screenshot of the Turbo C++ (TC) IDE. The main window has a dark blue background with yellow text. The menu bar at the top includes File, Edit, Run, Compile, Project, Options, Debug, and Break. The status bar at the top shows 'Line 10 Col 37 Insert Indent Tab Fill Unindent * E:2PM'. The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[10][10],nr,r,c,s=0;
clrscr();
printf("Enter no of rows ");scanf("%d",&nr);
printf("Enter %d integers\n",nr*nr);
for(r=0;r<nr;r++)for(c=0;c<nr;c++)
{ scanf("%d",&a[r][c]); if(r+c==nr-1)s+=a[r][c];}
printf("Trace=%d",s);
getch();
}
```

Below the code, there is a toolbar with icons for various functions. At the bottom of the IDE window, there is a status bar showing 'Page: 21 of 22 Words: 78' and a zoom level of '110%'. The Windows taskbar is visible at the very bottom, showing the Start button, taskbar icons for File Explorer, Calculator, DEV C++, ZOOM, Google Chrome, Word, Paint, and the IDE itself. The system clock shows '3:49 PM 28-Dec-24'.

```
TC
Enter no of rows 3
Enter 9 integers
1 0 8
4 0 9
1 2 3
Trace=9_

Activate Windows
Go to PC settings to activate Windows.

Page: 22 of 22 Words: 78 110% 3:49 PM 28-Dec-24
```

Trace: Sum of principle diagonal elements

if($r+c==nr-1$) $s+=a[r][c];$

1 0,0	2 0,1	3 0,2
4 1,0	5 1,1	9 1,2
5 2,0	7 2,1	8 2,2

trace =13

